

(*) All the rings should be regarded as commutative rings with identity. In this section, M is an A -module.

Recall the construction of the tensor product of two A -modules M and N :

Let $F = \bigoplus^{M \times N} A \triangleq A^{\oplus(M \times N)}$ (i.e., take as many copies of A as the number of the elements of $M \times N$). For $\forall (x, y) \in M \times N$, $(x, y) \mapsto e_{(x, y)} \in A^{\oplus(M \times N)}$, then $F = \langle e_{(x, y)} | (x, y) \in M \times N \rangle$ (for example, if $M = \mathbb{Z}_2$ and $N = \mathbb{Z}_3$, then $F = \bigoplus^{M \times N} A = \langle e_{00}, e_{10}, e_{01}, e_{11}, e_{02}, e_{12} \rangle = A^{(6)}$).

Let D be the submodule of F generated by the following elements:

$$e_{(x_1+x_2, y)} - e_{(x_1, y)} - e_{(x_2, y)}$$

$$e_{(x, y_1+y_2)} - e_{(x, y_1)} - e_{(x, y_2)}$$

$$e_{(ax, y)} - ae_{(x, y)}$$

$$e_{(x, ay)} - ae_{(x, y)}$$

here $\forall x_1, x_2, x \in M$, $\forall y_1, y_2, y \in N$ and $\forall a \in A$.

For any element $m \in F$, m is of the form $m \triangleq \sum_{i=1}^n a_i e_{(x_i, y_i)}$ and we have a correspondence

$$M \times N \hookrightarrow F$$

$$\sum_{i=1}^n a_i (x_i, y_i) \rightsquigarrow e_{\sum_{i=1}^n a_i (x_i, y_i)}$$

Now define $T := F/D$. Suppose P is any other A -module and $f : M \times N \rightarrow P$ is an A -bilinear map. Then f induces a map

$$\tilde{f} : F \rightarrow P$$

$$e_{(x, y)} \mapsto f(x, y)$$

it's easy to check that $\tilde{f}$ is an A -bilinear map.

$$\begin{array}{ccc} M \times N & \xrightarrow{f} & P \\ \downarrow & \nearrow \tilde{f} & \uparrow f' \\ F & \xrightarrow{\pi} & T = F/D \end{array}$$

Recall that for the diagram of modules:

$$\begin{array}{ccc} M & \xrightarrow{f} & N \\ \downarrow \pi & \nearrow h & \\ M/L & & \end{array}$$

if $L \subseteq \ker f$, then h can be well-defined as $h(m + L) := f(m)$ and h is unique.

By the definition of D , for the element $e_{(x_1+x_2,y)} - e_{(x_1,y)} - e_{(x_2,y)} \in F$, we have $\tilde{f}(e_{(x_1+x_2,y)} - e_{(x_1,y)} - e_{(x_2,y)}) = f(x_1+x_2, y) - f(x_1, y) - f(x_2, y) = 0$ (since they are bilinear). Thus for $\forall m \in D$, we have $\tilde{f}(m) = 0$, which means that $D \subseteq \tilde{f}$ and hence $\tilde{f}$ induces a map

$$f' : T = F/D \rightarrow P$$

such that $f'(\bar{e}_{(x,y)}) = \tilde{f}(e_{(x,y)}) = f(x, y)$.

Now we denote $\bar{e}_{(x,y)} = x \otimes y$ and define

$$g : M \times N \rightarrow T = F/D$$

$$(x, y) \mapsto x \otimes y$$

then we can know that g is an A -bilinear map since $g : (x_1 + x_2, y) \mapsto (x_1 + x_2) \otimes y = \bar{e}_{(x_1+x_2,y)} = \bar{e}_{(x_1,y)} + \bar{e}_{(x_2,y)} = x_1 \otimes y + x_2 \otimes y$ and we have $f' \circ g = \tilde{f}$.

$$\begin{array}{ccc} F & \xrightarrow{\tilde{f}} & P \\ g \downarrow & \nearrow f' & \\ T = F/D & & \end{array}$$

Now suppose there is an A -module T' and a map $g' : M \times N \rightarrow T'$ is A -bilinear such that given any A -bilinear map $f : M \times N \rightarrow P$, where P is an A -module, there exists a bilinear map $f' : T' \rightarrow P$ satisfying $f' \circ g' = f$, then $T' \cong T$. Indeed, let $P = T = F/D$. By the discussion above, we know that there exists a homomorphism $j : T' \rightarrow T$ such that $j \circ g' = g$,

$$\begin{array}{ccc} M \times N & \xrightarrow{g} & T \\ g' \downarrow & \nearrow j & \\ T' & & \end{array}$$

$$\begin{array}{ccc} M \times N & \xrightarrow{g} & T \\ g' \downarrow & \nearrow j' & \\ T' & & \end{array}$$

by the discussion above, $\exists j : T' \rightarrow T$ such that $j \circ g' = g$. By the given condition, $\exists j' : T' \rightarrow T$ such that $j' \circ g' = g$. Thus $j \circ j' \circ g' = g'$ and $j' \circ j \circ g = g$. Hence $j \circ j' = Id_{T'}$ and $j' \circ j = Id_T$, which means $T \cong T'$.

Hence we have the following proposition

Proposition 0.1. : Let M and N be two A -modules, then there exists an A -module T and an A -bilinear map $g : M \times N \rightarrow T$ such that for any A -module P and any A -bilinear map $f : M \times N \rightarrow P$, there exists an A -module homomorphism $f' : T \rightarrow P$ such that $f' \circ g = f$.

$$\begin{array}{ccc} M \times N & \xrightarrow{f} & P \\ \downarrow g & \nearrow f' & \\ T & & \end{array}$$

Further more, if there exists another A -module T' and an A -bilinear map g' satisfying the same property as above, then $T' \cong T$. (i.e., the module T is uniquely determined up to isomorphism)

Tensor Product of Modules

Definition 0.1. : The module T in proposition above is called the **tensor product of M and N** , denoted by $T = M \otimes N$.

Remark 0.1. :By the construction of tensor product, we can know that:

(1) $M \otimes N = \{\sum_{i=1}^n a_i(x_i \otimes y_i) | a_i \in A, x_i \in M, y_i \in N, n \in \mathbb{N}\}$. If $\{x_i\}_{i \in I}$ and $\{y_j\}_{j \in J}$ are generating sets for M and N respectively, then $\{x_i \otimes y_j\}_{i \in I, j \in J}$ is a generating set for $M \otimes N$. In particular, if M and N are both finitely generated, then so is $M \otimes N$.

(2) If M' and N' are submodules of M and N respectively, then $M' \otimes N'$ is a submodule of $M \otimes N$.

(3) By similar process, we can construct the tensor product of finitely many A -modules.

Example 0.1. :Let $A = M = \mathbb{Z}$ and $N = \mathbb{Z}_2$. Then for $2 \otimes \bar{1} \in M \otimes N = \mathbb{Z} \otimes \mathbb{Z}_2$, we have $2 \otimes \bar{1} = 2(1 \otimes \bar{1}) = 1 \otimes (2 \cdot \bar{1}) = 1 \otimes \bar{2} = 1 \otimes \bar{0} = 0$. (since for $\forall x \in M$, we have $x \otimes 0 = 0 \in M \otimes N$)

Example 0.2. :For $m, n \in \mathbb{N}$, if $\gcd(m, n) = 1$, then $\mathbb{Z}_n \otimes \mathbb{Z}_m = 0$.

Proof: For $\forall \bar{x} \otimes \bar{y} \in \mathbb{Z}_n \otimes \mathbb{Z}_m$, since $\gcd(m, n) = 1$, $\exists p, q \in \mathbb{Z}$ such that $1 = pn + qm$. So we have $\bar{x} \otimes \bar{y} = x(\bar{1} \otimes \bar{y}) = x(\overline{pn + qm} \otimes \bar{y}) = x(\overline{pn} \otimes \bar{y}) + x(\overline{qm} \otimes \bar{y}) = \overline{n(px)} \otimes \bar{y} + \bar{xq} \otimes \overline{my} = 0 + 0 = 0$. $\square$

Corollary 0.1. : If $\sum_{i=1}^n a_i(x_i \otimes y_i) = 0 \in M \otimes N$, then there exists submodules M_0 of M and N_0 of N such that $\sum_{i=1}^n a_i(x_i \otimes y_i) \in M_0 \otimes N_0$.

Proof: By the construction of $M \otimes N$, $\sum_{i=1}^n a_i(x_i \otimes y_i) = 0 \in M \otimes N \Leftrightarrow \sum_{i=1}^n a_i(x_i \otimes y_i) \in D$. Replacing the subscripts of the elements of the generating set of D by x_i and y_i respectively, we get M_0 and N_0 . $\square$

Proposition 0.2. :For the A -module M , we have $A \otimes M \cong M$ and $M \otimes A \cong M$.

Proof: Define

$$\varphi : A \times M \rightarrow M$$

$$(a, m) \mapsto am$$

then φ is an A -bilinear map. Therefore, there exists an A -bilinear map $\varphi' : A \otimes M \rightarrow M$ such that $\varphi'(a \otimes m) = \varphi(a, m) = am$.

There exists a natural map

$$\psi : M \rightarrow A \otimes M$$

$$m \mapsto 1 \otimes m$$

Since an element $x \in A \otimes M$ is of the form

$$x \triangleq \sum_{i=1}^n a_i \otimes m_i = \sum_{i=1}^n 1 \otimes a_i m_i = 1 \otimes \left(\sum_{i=1}^n a_i m_i \right) \triangleq 1 \otimes m,$$

therefore $\varphi' \circ \psi = Id_M$ and $\psi \circ \varphi' = Id_{A \otimes M}$ and $A \otimes M \cong M$. $\square$

Proposition 0.3. : Let M be an A -module and I is an ideal of A , then $A/I \otimes M \cong M/IM$.

Proof: Consider the natural homomorphism

$$\varphi : M \rightarrow A/I \otimes M$$

$$m \mapsto \bar{1} \otimes m$$

then for $\forall x \in IM$, x is of the form $x \triangleq \sum_{i=1}^n a_i m_i$ where $a_i \in I$ and $m_i \in M$. Thus $\varphi(x) = \bar{1} \otimes (\sum_{i=1}^n a_i m_i) = \sum_{i=1}^n \bar{a}_i \otimes m_i = \sum_{i=1}^n \bar{0} \otimes m_i$ (note that $a_i \in I$), which means that $\varphi(x) = 0$ and $IM \subseteq \ker \varphi$. Hence we have the diagram

$$\begin{array}{ccc} M & \xrightarrow{\varphi} & A/I \otimes M \\ \downarrow & \nearrow \bar{\varphi} & \\ M/IM & & \end{array}$$

i.e., $\varphi : M \rightarrow A/I \otimes M$ filters through M/IM and

$$\bar{\varphi} : M/IM \rightarrow A/I \otimes M$$

$$\bar{m} \mapsto \varphi(m) = \bar{1} \otimes m$$

$\bar{\varphi}$ can be checked that it is well-defined. Now, a natural map can be given

$$\bar{\psi} : A/I \otimes M \rightarrow M/IM$$

$$\bar{a} \otimes m \mapsto \overline{am}$$

note that $\bar{\psi}$ is clearly well-defined if we consider the bilinear map $\psi : A/I \times M \rightarrow M/IM$, $(\bar{a}, m) \mapsto \overline{am}$.

It is easy to check that $\bar{\varphi} \circ \bar{\psi} = Id_{A/I \otimes M}$ and $\bar{\psi} \circ \bar{\varphi} = Id_{M/IM}$ and hence $A/I \otimes M \cong M/IM$. $\square$

Here are some other properties of tensor product

Proposition 0.4. :Let M, N, P be A -modules. Then there exists some natural isomorphism as follows:

(1)

$$M \otimes N \xrightarrow{\cong} N \otimes M$$

$$m \otimes n \longmapsto n \otimes m$$

(2)

$$(M \otimes N) \otimes P \xrightarrow{\cong} M \otimes (N \otimes P) \xrightarrow{\cong} M \otimes N \otimes P$$

$$(m \otimes n) \otimes p \longmapsto m \otimes (n \otimes p) \longmapsto m \otimes n \otimes p$$

(3)

$$(M \oplus N) \otimes P \xrightarrow{\cong} (M \otimes P) \oplus (N \otimes P)$$

$$(m, n) \otimes p \longmapsto (m \otimes p, n \otimes p)$$

Let $f : M \rightarrow M'$ and $g : N \rightarrow N'$ be A -module homomorphisms, then there is an A -bilinear map

$$(f, g) : M \times N \rightarrow M' \otimes N'$$

$$(m, n) \mapsto f(m) \otimes g(n)$$

by the construction of tensor product, we can know that there exists an A -module homomorphism, denoted by $f \otimes g$,

$$f \otimes g : M \otimes N \rightarrow M' \otimes N'$$

such that $(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$, i.e, the diagram commutes

$$\begin{array}{ccc} M \times N & \xrightarrow{(f, g)} & M' \otimes N' \\ \otimes \downarrow & \nearrow f \otimes g & \\ M \otimes N & & \end{array}$$

Further more, for the module homomorphisms

$$M \xrightarrow{f} M' \xrightarrow{f'} M''$$

$$N \xrightarrow{g} N' \xrightarrow{g'} N''$$

we have the compositions

$$M \xrightarrow{f' \circ f} M''$$

$$N \xrightarrow{g' \circ g} N''$$

then, there exists a homomorphism

$$(f' \circ f) \otimes (g' \circ g) : M \otimes N \rightarrow M'' \otimes N'',$$

on the other hand, we have the map

$$(f' \otimes g') \circ (f \otimes g) : M \otimes N \rightarrow M' \otimes N' \rightarrow M'' \otimes N''$$

then for $\forall m \otimes n \in M \otimes N$, $(f' \circ f) \otimes (g' \circ g)(m \otimes n) = (f' \circ f)(m) \otimes (g' \circ g)(n) = (f' \otimes g')(f(m) \otimes g(n)) = (f' \otimes g')((f \otimes g)(m \otimes n)) = [(f' \otimes g') \circ (f \otimes g)](m \otimes n)$, which means that

Proposition 0.5. :For the module homomorphisms

$$M \xrightarrow{f} M' \xrightarrow{f'} M''$$

$$N \xrightarrow{g} N' \xrightarrow{g'} N''$$

we have

$$(f' \circ f) \otimes (g' \circ g) = (f' \otimes g') \circ (f \otimes g) : M \otimes N \rightarrow M'' \otimes N''$$

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Let M, N, P be A -modules, given an A -bilinear map $\varphi : M \times N \rightarrow P$, then for any $x \in M$, we have an A -module homomorphism

$$M \rightarrow \text{Hom}_A(N, P)$$

$$x \mapsto f(x, -)$$

where

$$f(x, -) : N \rightarrow P$$

$$n \mapsto f(x, -)(n) := \varphi(x, n) \quad (\forall n \in N)$$

Now for any $f \in \text{Hom}_A(M, \text{Hom}_A(N, P))$, then f induces an A -bilinear map from $M \times N$ to P given by

$$\bar{f} : M \times N \rightarrow P$$

$$(m, n) \mapsto (f(m))(n)$$

Define

$$\varphi' : \text{Hom}_A(M, \text{Hom}_A(N, P)) \rightarrow \text{Hom}_A(M \otimes N, P)$$

$$f \mapsto \varphi'(f)$$

where $(\varphi'(f))(m \otimes n) := (f(m))(n)$. Then we can check that φ' is an isomorphism.

Indeed, for $\forall g \in \ker \varphi'$, then we have $\varphi'(g) = 0 : M \otimes N \rightarrow P$, hence $\varphi'(g)(m \otimes n) = (g(m))(n) = 0$ ($\forall m, n$), therefore $g(m) = 0$ ($\forall m$) and $g = 0 \in \text{Hom}_A(M, \text{Hom}_A(N, P))$, which implies that φ' is injective;

On the other hand, for $\forall g \in \text{Hom}_A(M \otimes N, P)$, we need to find a map $\alpha \in \text{Hom}_A(M, \text{Hom}_A(N, P))$ such that $\varphi'(\alpha) = g$. Define

$$h : M \times N \rightarrow P$$

$$(m, n)m \mapsto g(m \otimes n)$$

and define $\alpha(m) := h(m, -)$ ($\forall m \in M$), where $h(m, -)(n) = h(m, n)$. Then it's easy to check that $\varphi'(\alpha) = g$ and hence φ' is surjective.

Hence we have the following proposition

Proposition 0.6. :For the A -modules M, N and P , we have

$$\text{Hom}_A(M, \text{Hom}_A(N, P)) \cong \text{Hom}_A(M \otimes N, P)$$

.

Tensor Product is Right Exact

Proposition 0.7. :If

$$M' \xrightarrow{f} M \xrightarrow{g} M'' \longrightarrow 0$$

is an exact sequence of A -modules, then so is

$$M' \otimes N \xrightarrow{f \otimes 1} M \otimes N \xrightarrow{g \otimes 1} M'' \otimes N \longrightarrow 0$$

for any A -module N .

Proof: Since

$$M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is an exact sequence, by the left exactness of hom functor, we have the exact sequence

$$0 \longrightarrow \text{Hom}_A(M'', \text{Hom}_A(N, P)) \longrightarrow \text{Hom}_A(M, \text{Hom}_A(N, P)) \longrightarrow \text{Hom}_A(M', \text{Hom}_A(N, P))$$

by proposition 0.6, we know that

$$0 \longrightarrow \text{Hom}_A(M'' \otimes N, P) \longrightarrow \text{Hom}_A(M \otimes N, P) \longrightarrow \text{Hom}_A(M' \otimes N, P)$$

is an exact sequence. Therefore,

$$M' \otimes N \longrightarrow M \otimes N \longrightarrow M'' \otimes N \longrightarrow 0$$

is an exact sequence. $\square$

Example 0.3. :Consider the exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z}$$

take another $\mathbb{Z}$ -module $\mathbb{Z}_2$. Then

$$0 \longrightarrow \mathbb{Z} \otimes \mathbb{Z}_2 \xrightarrow{2 \otimes 1} \mathbb{Z} \otimes \mathbb{Z}_2$$

is not exact since

$$(2 \otimes 1)(n \otimes \bar{1}) = (2n) \otimes \bar{1} = n \otimes (2 \cdot \bar{1}) = 0$$

which means that $2 \otimes 1$ is not injective. So tensor product is not left exact.

Flat Module

Definition 0.2. :For any exact sequence of A -modules

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

if there exists an A -module N such that

$$0 \longrightarrow M' \otimes N \longrightarrow M \otimes N \longrightarrow M'' \otimes N \longrightarrow 0$$

is exact, then N is called a **flat A -module**.

We have some properties about flat modules

Proposition 0.8. : For an A -module N , the following are equivalent:

(1) N is flat;

(2)

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is an exact sequence implies

$$0 \longrightarrow M' \otimes N \longrightarrow M \otimes N \longrightarrow M'' \otimes N \longrightarrow 0$$

is exact;

(3) $f : M' \rightarrow M$ is injective implies that $f \otimes 1 : M' \otimes N \rightarrow M \otimes N$ is injective;

(4) For any finitely generated A -modules M' and M , $f : M' \rightarrow M$ is injective implies that $f \otimes 1 : M' \otimes N \rightarrow M \otimes N$ is injective.

Proof: (1) $\Leftrightarrow$ (3): By definition;

(2) $\Leftrightarrow$ (3):By the right exactness of tensor product;

(3) $\Rightarrow$ (4):Trivial;

(4) $\Rightarrow$ (3): Let $f : M' \rightarrow M$ be injective and take $\forall u \triangleq \sum_{i=1}^n x_i \otimes y_i \in \ker(f \otimes 1)$, then $(f \otimes 1)(u) = \sum_{i=1}^n f(x_i) \otimes y_i = 0$. Let M'_0 be the submodule of M' generated by

$\{x_1, x_2, \dots, x_n\}$. Since $\sum_{i=1}^n f(x_i) \otimes y_i = 0$, so there exists a finitely generated submodule M_0 of M containing $f(M'_0)$ such that $\sum_{i=1}^n f(x_i) \otimes y_i = 0 \in M_0 \otimes N$, let $u_0 = \sum_{i=1}^n x_i \otimes y_i$ be the element of $M'_0 \otimes N$, then $f|_{M'_0} : M'_0 \rightarrow M_0$ is injective. By the given condition, we know that $f|_{M'_0} \otimes 1 : M'_0 \otimes N \rightarrow M_0 \otimes N$ is injective. Hence if $(f|_{M'_0} \otimes 1)(u_0) = 0$, then we have $u_0 = 0 \in M'_0 \otimes N$ and therefore $u = 0 \in M' \otimes N$, which implies $\ker(f \otimes 1) = \{0\}$ and $f \otimes 1$ is injective. $\square$